CO2 ASSESSMENT TOOL -1

MLA0202-FUNDAMENTALS OF MACHINE LEARNING

Analytical Problem Solving

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I K.Hemakshitha(192425088) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.I understand that any violation of academic integrity rules will result in disciplinary action**.**

Signature of the student:

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QUESTION 1:

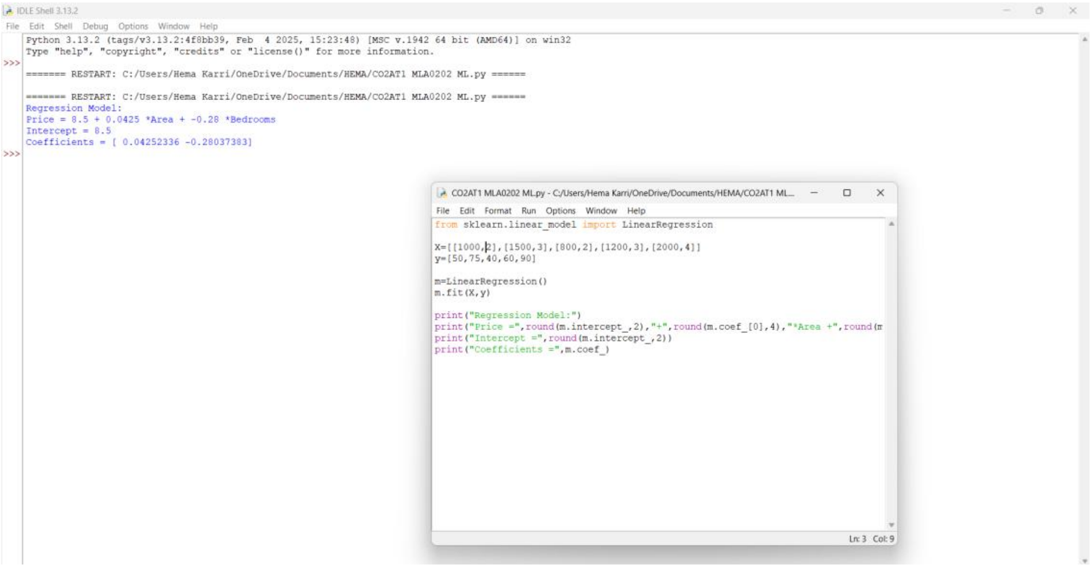
CODE:

```
from sklearn.linear_model import LinearRegression


X=[[1000,2],[1500,3],[800,2],[1200,3],[2000,4]]

y=[50,75,40,60,90]


m=LinearRegression()

m.fit(X,y)


print("Regression Model:")

print("Price =",round(m.intercept_,2),"+",round(m.coef_[0],4),"*Area
+",round(m.coef_[1],2),"*Bedrooms")
```

print("Intercept =",round(m.intercept_,2))

print("Coefficients =",m.coef_)

OUTPUT:



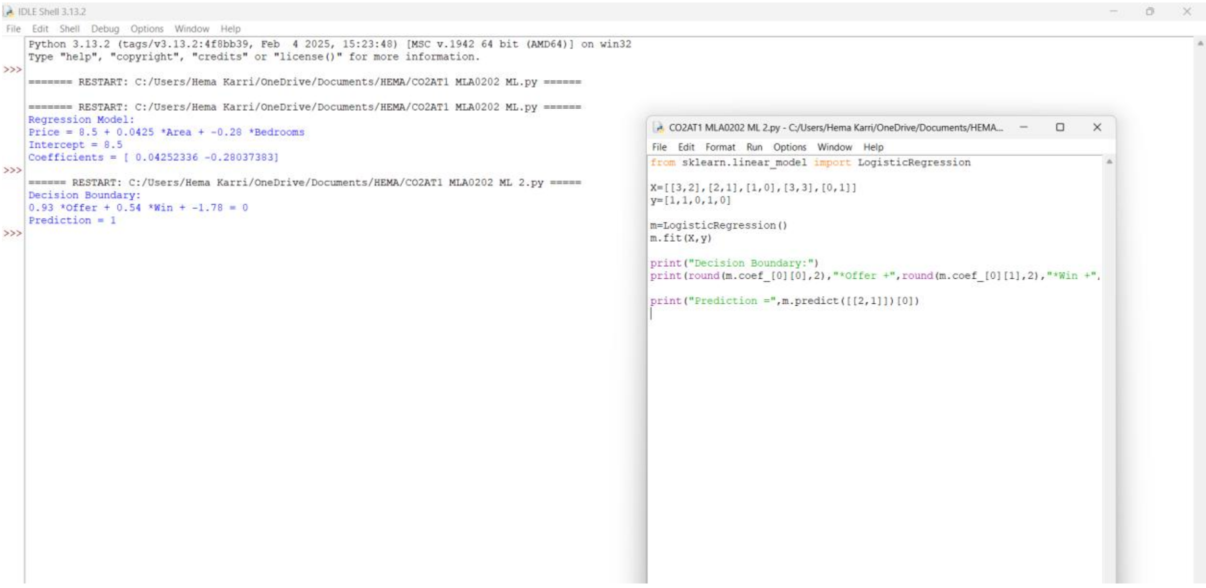
QUESTION 2:

CODE:

```
from sklearn.linear_model import LogisticRegression

X=[[3,2],[2,1],[1,0],[3,3],[0,1]]
y=[1,1,0,1,0]

m=LogisticRegression()
m.fit(X,y)

print("Decision Boundary:")
print(round(m.coef_[0][0],2),"*Offer +",round(m.coef_[0][1],2),"*Win +",round(m.intercept_[0],2),"= 0")
```

```
print("Prediction =",m.predict([[2,1]])[0])
```

OUTPUT:



QUESTION 3:

CODE:

```
py,pn=3/5,2/5
pfy,phy=3/3,1/3
pfn,phn=0/2,1/2

yes=py*pfy*phy
no=pn*pfn*phn

print("P(Yes) =",py)
print("P(No) =",pn)
print("P(Fever=1|Yes) =",pfy)
print("P(Headache=0|Yes) =",phy)
```

```
print("Probability(Yes) =",yes)
print("Probability(No) =",no)
print("Disease =","Yes" if yes>no else "No")
```

OUTPUT: